

Prioritizing Causal Genes at GWAS Loci

Extending PoPS with genomic foundation models

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Genetic Concepts and GWAS

From variants to traits

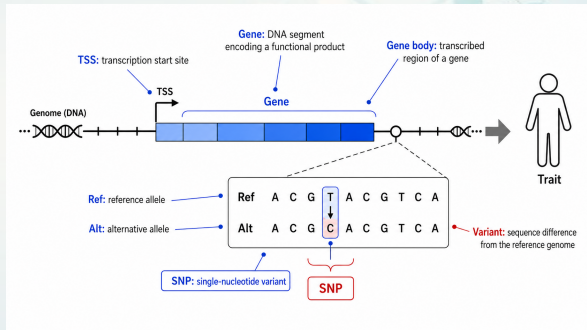
Single-nucleotide polymorphisms (**SNPs**) are common single-base differences between individual genomes in a population.

Some SNPs may be associated with traits such as height, asthma, or blood biomarkers.

GWAS [NHGRI GWAS]

Genome-wide association studies systematically test associations between genetic variants and traits.

Because nearby variants are correlated, a GWAS hit often identifies a **locus**, not the exact causal gene.



MAGMA [de Leeuw et al., 2015]

MAGMA is a gene-level association method. It combines SNP-level GWAS signals within a gene while accounting for linkage disequilibrium. The result is a **gene association score** that tells us how strongly each gene is linked to the trait genome-wide. These scores are then treated as baseline to derive PoPS.

PoPS [Weeks et al., 2023]

PoPS (Polygenic Priority Score) uses genome-wide gene features such as expression, pathways, and protein interactions to learn what properties are shared by trait-relevant genes. It then scores genes in a locus by how well they match that learned trait-specific profile.

How they work together

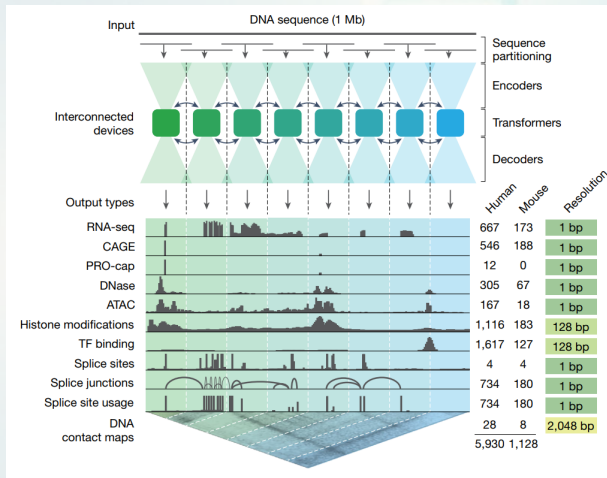
MAGMA gives the target signal for each gene, while PoPS uses biological features to explain and prioritize that signal with a regularized linear model. In our project, PoPS provides the baseline gene-ranking framework we extend with sequence-based features.

AlphaGenome [Žiga Avsec et al. 2026]

Input: DNA sequence of up to approximately 1 Mb.

Architecture: Sequence encoders, interconnected transformers, and task-specific decoders model local and long-range genomic regulation.

Output: Thousands of functional predictions across cell types: gene expression, chromatin state, TF binding, RNA splicing, chromatin contacts. We use its Δ RNA (RNA-abundance) variant effects.



AlphaGenome maps long DNA sequences to multiple functional genomic readouts through encoder, transformer, and decoder modules.

Problem Statement

Formal Task

Genes $j \in \mathcal{G}$. Learn a scoring map $s : \mathbb{R}^d \rightarrow \mathbb{R}$ so that within each locus L the top-scored gene is the causal one:

$$\hat{g}_L = \arg \max_{j \in \mathcal{G}_L} s(\mathbf{f}_j), \quad \text{Acc} = \frac{1}{|\mathcal{L}|} \sum_{L \in \mathcal{L}} \mathbb{1}[\hat{g}_L = g_L^*].$$

Feature maps $\mathbf{f}_j$ (per gene j):

$$\underbrace{\mathbf{x}_j^{\text{PoPS}} \in \mathbb{R}^{57742}}_{\text{PoPS gene features}}, \quad \underbrace{\mathbf{x}_j^{\text{AG}} \in \mathbb{R}^{285}}_{\text{AlphaGenome } \Delta\text{RNA, 285 ontologies}}, \quad \underbrace{y_j = \text{ZSTAT}_j \in \mathbb{R}}_{\text{MAGMA Z-statistic (target)}}.$$

Gold standard [Finucane Lab data]

A dataset of 1,348 loci (each with a locus ID, a credible-set ID, and a trait). Every locus contains a fine-mapped **coding** variant, and the gene it sits in is **taken as** the causal gene g_L^* .

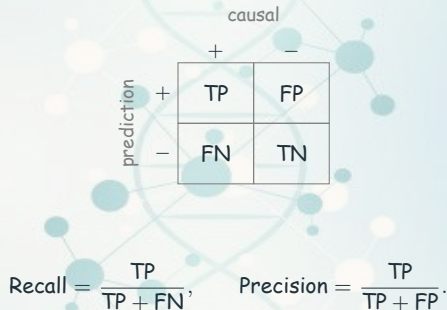
where: $\mathcal{G}$ = gene set ($|\mathcal{G}|=15,795$ with PoPS features, 9,669 with AlphaGenome); $\mathbf{f}_j$ = the feature block fed to s , dimension $d = 58,027$; $\mathcal{G}_L \subset \mathcal{G}$ = candidate genes for locus L ; $\mathcal{L}$ = the 1,348 loci; $s(\cdot)$ = learned score; $\hat{g}_L$ = predicted gene; g_L^* = causal (labelled) gene; $\mathbb{1}[\cdot]$ = indicator; Acc = top-1-per-locus accuracy.

Data Dimensions

Dataset	Dimensions
PoPS Features	15,795 genes $\times$ 57,742 features
AlphaGenome Δ RNA	9,669 genes $\times$ 285 ontologies
MAGMA z_{STAT} scores	15,795 genes
PoPS Validation Set	1,348 loci

Method 1: The Validation Data Set

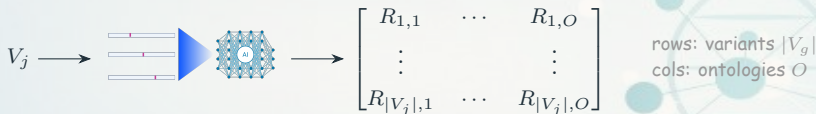
- PoPS incorporates orthogonal information into MAGMA using gene features.
- Assessed using precision and recall.
- We used a validation set encompassing 4,187 genes across 1348 loci and 95 traits from the PoPS paper.
 - Also included raw MAGMA scores for genes across each locus.
 - Marked probable causal genes using fine-mapping results as a comparison benchmark.



Building the gene $\times$ ontology feature matrix

1. **Variants near each gene.** For a gene j in the set of all genes G , consider all nucleotides within 500 kb of its TSS. This gives a set of variants V_j for each gene.

2. **Score variants with AlphaGenome.** Pass each $v \in V_j$ through AlphaGenome, recording the RNA-seq changes $R_{i,j}$ across O ontologies ($O = 285$).



3. **Collapse to one vector per gene.** Weighted mean over all variants, each weight an exponential decay over distance:

$$x_j^{AG} = \frac{1}{\sum_{i=1}^{|V_j|} w_i} \sum_{i=1}^{|V_j|} w_i [R_{i,1} \quad \cdots \quad R_{i,O}]$$

row i = variant i 's ontology vector

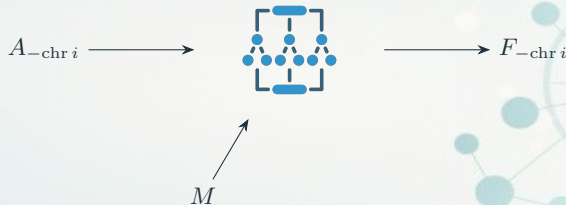
4. **Stack into a matrix.** Each aggregate gene vector becomes a row:

$$A = \begin{bmatrix} x_1^{AG} \\ \vdots \\ x_{|G|}^{AG} \end{bmatrix}$$

(# genes $\times$ # ontologies)

Deriving Model Scores

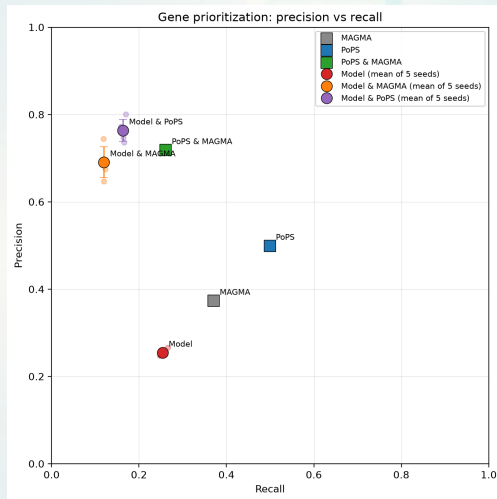
For each of the 22 chromosomes, fit a random forest $F_{\text{chr } i}$ (L. Breiman. 2001) that ignores all genes on that chromosome.



- M = vector of raw MAGMA scores for each gene (excluding $\text{chr } i$).
- $F_{\text{chr } i}$ is then used to predict a model score for the held-out chromosome.
- This allows us to rank each gene.

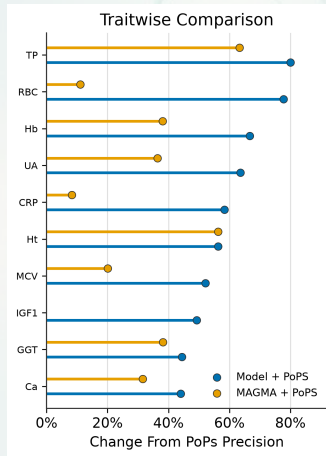
Method 1 results: Recall and Precision Comparison

- Alone, the model reaches $\sim 25\%$ recall and precision, underperforming MAGMA and PoPS.
- Intersecting our model's gene set with the PoPS/MAGMA sets raises precision to upwards of 77%.
- Indicative of a potential orthogonal signal



Method 1 results: Precision by Trait

- Across the top 10 traits at which the combination of our model and PoPS improved precision, we see that the Model+PoPS combination improved precision across traits differently than MAGMA+PoPS, especially for red blood cell count (RBC)
- The improvement pattern is different from MAGMA+PoPS, indicating AlphaGenome is potentially providing some orthogonal signal, improving precision in a unique context



Method 2: Two views, Two fusions

Two views

Learn one view of what a gene is from PoPS, and one view of what nearby variants do from AlphaGenome.

Two ways to combine them:

- **Late fusion (2a):** train each view separately, then fuse their held-out scores (ridge-penalised stacking).
- **Early fusion (2b):** concatenate the two feature sets, then fit a single ridge model.

Why ridge?

- **On PoPS/[PoPS+AG]:** A total of 57,742/58,027 features $\gg$ genes and are highly correlated — an ℓ_2 penalty controls variance/multicollinearity ($p \gg n$).
- **In stacking (late fusion):** the two block-scores are **positively correlated** since they are both designed to predict MAGMA ZSTAT, so ℓ_2 -penalised weights are shrunk rather than fit unconstrained.

Evaluation

Top-1 accuracy per locus on the 1,348 coding loci, compared trait-by-trait to the original PoPS baseline.

AlphaGenome aggregation: CADD- and distance-weighted sum

For gene j , let $\{i\}$ index SNPs within 1 Mb of the gene, each with an AlphaGenome Δ RNA vector $v_i \in \mathbb{R}^{285}$, a distance d_i to the gene body, and a CADD PHRED score C_i .

Weight and aggregate (weighted sum)

$$w_{ij} = \underbrace{\begin{cases} 1 & d_i = 0 \text{ (inside body)} \\ \exp(-d_i/\tau) & d_i > 0 \end{cases}}_{\text{distance}} \times \underbrace{(C_i/c)}_{\text{CADD (PHRED)}}, \quad \mathbf{x}_j^{\text{AG}} = \sum_i w_{ij} \mathbf{v}_i \in \mathbb{R}^{285}.$$

$$\tau = 100 \text{ kb}, \quad c = 1.0 \text{ (CADD scale)}, \quad d_i = \text{distance to [start, end]}.$$

- **CADD** scores each variant's predicted deleteriousness. The **PHRED** form C_i is a genome-wide rank score where higher C_i up-weights likely functional variants.
- Variants **inside the gene body** get full weight (1). Outside, weight decays with distance. More deleterious (high CADD) variants count more.
- Variants with missing CADD are dropped whereas a gene with no usable variant gets a mean-imputed (neutral) $\mathbf{x}_j^{\text{AG}}$.

Method 2: Ridge Regression

We predict the MAGMA target $y = \text{ZSTAT}$ from a feature matrix X (PoPS, AlphaGenome, or the two combined).

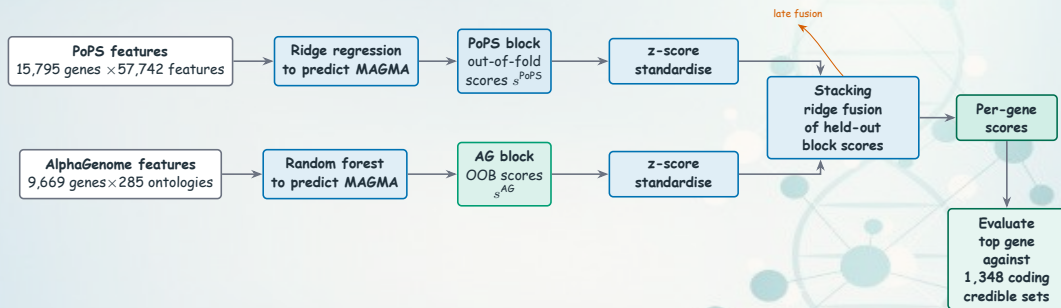
Ridge estimator

$$\hat{\beta}(\lambda) = \arg \min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 = (X^\top X + \lambda I)^{-1} X^\top y.$$

- The ℓ_2 penalty $\lambda \|\beta\|_2^2$ shrinks coefficients, stabilizing the fit when $p \gg n$ and features are correlated.
- λ is chosen by cross-validation.
- Scores are taken from **out-of-fold** predictions so a gene is never scored by a model that saw it, as this prevents leakage into the locus evaluation.

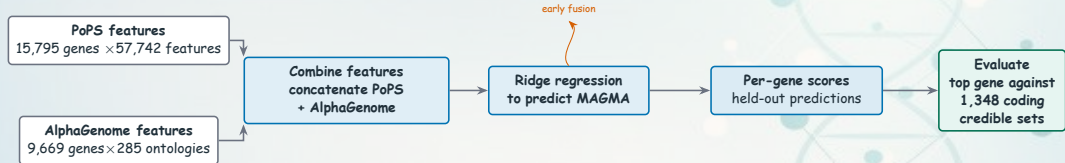
where: $y = \text{ZSTAT} \in \mathbb{R}^n$ = target over n genes; $X \in \mathbb{R}^{n \times p}$ = feature matrix ($p=57,742$ PoPS, 285 AG, or 58,027 combined); $\beta \in \mathbb{R}^p$ = coefficients; $\lambda \geq 0$ = ridge penalty; $I = p \times p$ identity; $\hat{\beta}(\lambda)$ = ridge solution; out-of-fold = Cross Validation-held-out.

Method 2a: Stacking (Late Fusion)



- **RF (AG), regularised for tractability:** 100 trees, 5% of features per split, ≥ 20 genes per leaf.
- **Held-out scoring:** out-of-bag (OOB), not leave-one-chromosome-out (LOCO) as causal-gene labels come from independent coding variants.
- **Fuse with ridge:** standardise s^{PoPS} , s^{AG} , then $\hat{y} = \gamma_0 + \gamma_1 z^{PoPS} + \gamma_2 z^{AG}$. This is because the two block-scores are correlated, so γ_1, γ_2 are ℓ_2 -shrunk rather than fit unconstrained.
- **Missing block:** genes in only one block use that block's mean for the absent score.

Method 2b: Ridge on Combined Features (Early Fusion)



Column-concatenate the feature blocks, $X^{\text{comb}} = [X^{\text{PoPS}} \parallel X^{\text{AG}}]$, then fit a single ridge model to learn one combined gene score. Genes outside AlphaGenome's 9,669 have their 285 AG entries **mean-imputed** (neutral after standardisation), so they keep all PoPS features and the score falls back to PoPS.

where: $X^{\text{PoPS}} \in \mathbb{R}^{15795 \times 57742}$, $X^{\text{AG}} \in \mathbb{R}^{15795 \times 285}$; $X^{\text{comb}} = [X^{\text{PoPS}} \parallel X^{\text{AG}}] \in \mathbb{R}^{15795 \times 58027}$;

Method 2 results: Trait-wise Precision

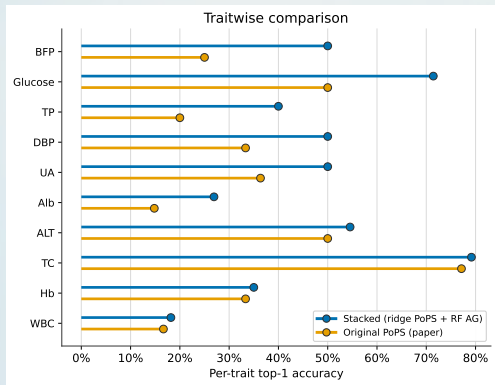


Figure 1: Trait-wise precision for the **stacked model** (ridge PoPS + RF AlphaGenome) versus the original PoPS baseline.

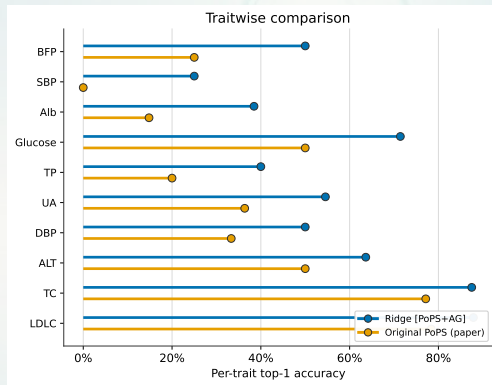


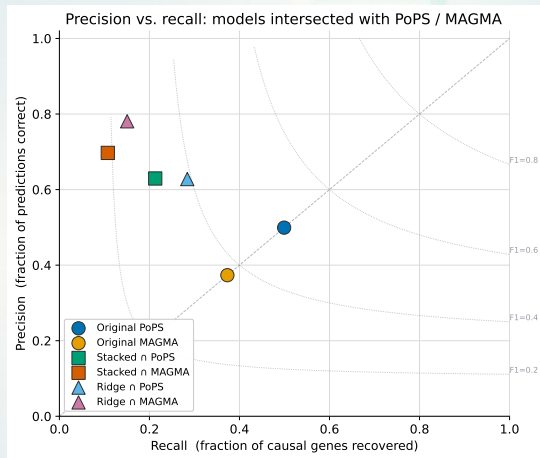
Figure 2: Trait-wise precision for **Ridge [PoPS+AG]** versus the original PoPS baseline.

Interpretation: Adding AlphaGenome to PoPS matches or exceeds the PoPS baseline on the above traits. The largest gains recur on blood-cell and metabolic traits, echoing the orthogonal signal seen in Method 1.

Method 2 results: intersecting with an orthogonal prior

Predict a gene **iff both** our model and a prior (PoPS or MAGMA) rank it top-1 in the credible set. Requiring agreement trades **recall** for **precision**.

- Baselines lie on the diagonal (precision = recall).
- **Ridge** $\cap$ **MAGMA**: the shared gene is causal **78.1%** of the time (vs **37.3%** for MAGMA alone), on the **19%** of sets where they agree.
- **Ridge** $\cap$ **PoPS**: this model probably provides the best balance with **precision 0.63** at **recall 0.28**.

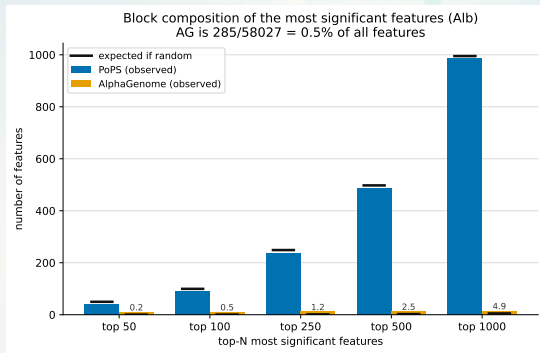


where: intersection (AND) rule: gene predicted for credible set L iff $\hat{g}_L^{\text{model}} = \hat{g}_L^{\text{prior}}$; precision = correct/committed; recall = correct/1,348; committed = sets where model and prior agree.

Which features matter? AlphaGenome punches above its weight

Ridge spreads weight across correlated features, so we rank importance with a **sparse** model (a marginal screen followed by an elastic net). For **Albumin** (a biomarker where Ridge [PoPS+AG] beats PoPS):

- The **4 most significant features are all AlphaGenome**.
- 10 of the top 50 are AlphaGenome, a **$\sim 41\times$** over-representation (0.25 expected by chance).
- 13 of 159 selected features are AlphaGenome, a **$\sim 17\times$** over-representation (8.2% vs 0.5%).



AlphaGenome barely moves pooled accuracy, but on biomarker traits its features carry the **largest** weights.

where: expected count if random = $N \times 285/58,027$; AlphaGenome = 285 of 58,027 features (0.5%); "top-N" ranked by |elastic-net coefficient|.

Feature importance: AlphaGenome tissue ontologies dominate the top

Of the **top 50** Albumin features (elastic net ranked by $|\text{coef}|$), **7 of the 10 are uberon** (tissue/anatomy) ontologies, only 2 are cell-type (cl) and 1 experimental-factor (efo). This suggests that the sequence signal enters through tissue-level regulatory context, not single cell types. The tissues are diverse (brain, heart, kidney), so this is a **block-level** statistical signal.

Rank	Ontology	Type
1	UBERON:0000955 (brain)	tissue
2	CL:0002319 (neural cell)	cell
3	UBERON:0018118	tissue
4	UBERON:0000948 (heart)	tissue
14	UBERON:0004538	tissue
17	EFO:0030018	exp. factor
20	CL:0000047	cell
31	UBERON:0001072	tissue
33	UBERON:0008971	tissue
49	UBERON:0015143	tissue

The 10 AlphaGenome features among the top 50.

where: uberon = anatomy/tissue ontology; cl = cell ontology; efo = experimental factor ontology.

Some Observations

- **Model class, not features, was decisive:** ridge on PoPS nearly doubled the random forest on identical features (0.44 vs 0.24), recovering $\sim 87\%$ of PoPS.
- **Pooled, AlphaGenome is largely redundant** once a strong linear model has the PoPS features (sees improvement by $+0.002$).
- **But the signal is real and trait-specific:** on serum biomarkers (total protein, cholesterol, urate, albumin, SHBG) our models exceed PoPS, and the stacking weights independently load on AlphaGenome for lipid, blood-cell and endocrine traits.
- **Intersection buys precision:** when our model and MAGMA agree, the gene is causal 78% of the time, although at the cost of recall.
- **Integration strategy matters:** late fusion beat early fusion, and un-scaled concatenation was actively harmful.

Limitations & Future Work

Limitations

- AlphaGenome features are restricted to **RNA-abundance** effects only, likely a cause of the low recall.
- AlphaGenome covers fewer genes (9,669) than PoPS (15,795), shrinking the overlap for early fusion.
- We have not yet fully replicated the original PoPS pipeline locally due to time constraints and lack of availability of data, causing the precision to decrease.

Future work

- Replicate the original PoPS framework on combined PoPS + AlphaGenome features.
- Extend AlphaGenome beyond RNA to splicing, chromatin state, and TF binding.
- Integrate AlphaGenome features with cS2G SNP-to-gene linking strategies.

Thank You!

Questions & Discussion

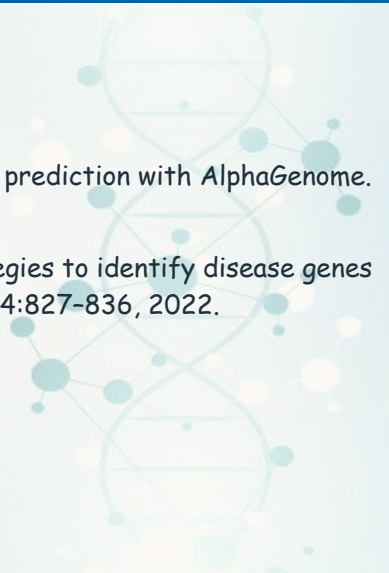
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References i

-  C. A. de Leeuw et al. *MAGMA: generalized gene-set analysis of GWAS data*. *PLoS Comput. Biol.* 11(4):e1004219, 2015.
-  E. M. Weeks et al. *Leveraging polygenic enrichments of gene features to predict genes underlying complex traits and diseases*. *Nature Genetics* 55:1267-1276, 2023.
-  L. Breiman. *Random forests*. *Machine Learning* 45(1):5-32, 2001.
-  P. Rentzsch et al. *CADD: predicting the deleteriousness of variants throughout the human genome*. *Nucleic Acids Res.* 47(D1):D886-D894, 2019.
-  R. J. Klein et al. *Complement factor H polymorphism in age-related macular degeneration*. *Science* 308:385-389, 2005.

- 
-  Ž. Avsec et al. Advancing regulatory variant effect prediction with AlphaGenome. Nature, 2026 (bioRxiv 2025.06.25.661532).
 -  S. Gazal et al. Combining SNP-to-gene linking strategies to identify disease genes and assess disease omnigenicity. Nature Genetics 54:827-836, 2022.